Equilibrium

Answer key

Conventions used throughout

• **Forces:** F_g = gravity (F_g = mg), F_N = normal force, F_T = tension, F_k = kinetic friction, F_{air} = air resistance. A subscript x or y means that force's x or y component, so F_{gx} is the x component of gravity.

• **Method (from the lesson):** draw the FBD, choose + and − for x and y, split angled forces into components, then use $\Sigma F_x = ma_x$ and $\Sigma F_y = ma_y$. Equilibrium means a = 0, so $\Sigma F_x = \Sigma F_y = 0$.

• **Signs in the working:** plain + and − numbers. **Directions in final answers:** [E], [N 14° W], and so on.

• g = 9.81 m/s^2. Every step keeps full precision; only final answers are rounded.

EXAMPLE 1 (FBDS)

Free-body diagrams

Example 1) Draw a free- body diagram for the following scenarios:

a) A skydiver is descending with a constant velocity (consider air resistance).

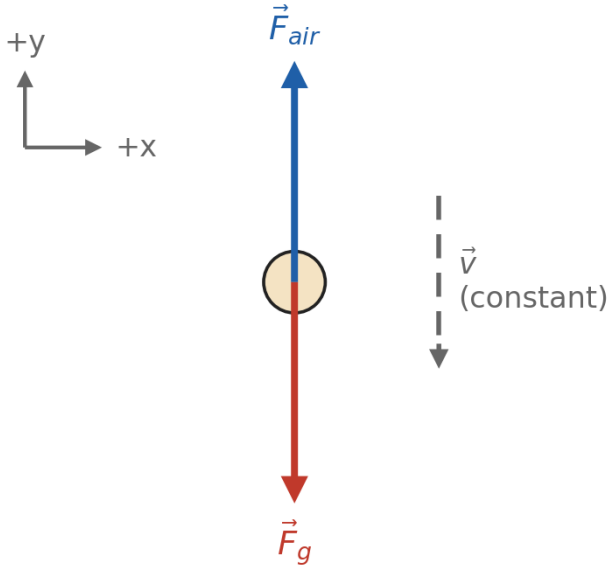
b) A toboggan sliding down the hill.

c) A block getting pulled up a ramp by a string.

(a) A skydiver is descending with a constant velocity (consider air resistance).

Constant velocity means a = 0, so the net force is zero. Air resistance (up) must be exactly as big as gravity (down): draw the two arrows the same length.

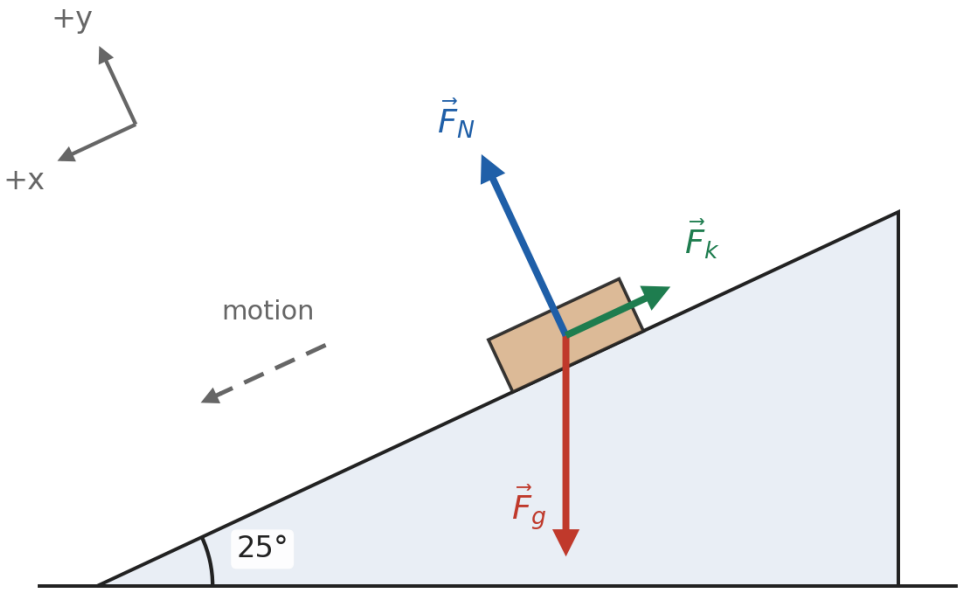
FBD: skydiver



(b) A toboggan sliding down the hill.

Three forces: gravity straight down, the normal force perpendicular to the hill, and friction from the snow up the hill (opposite the motion). +x is down the hill, the direction it moves.

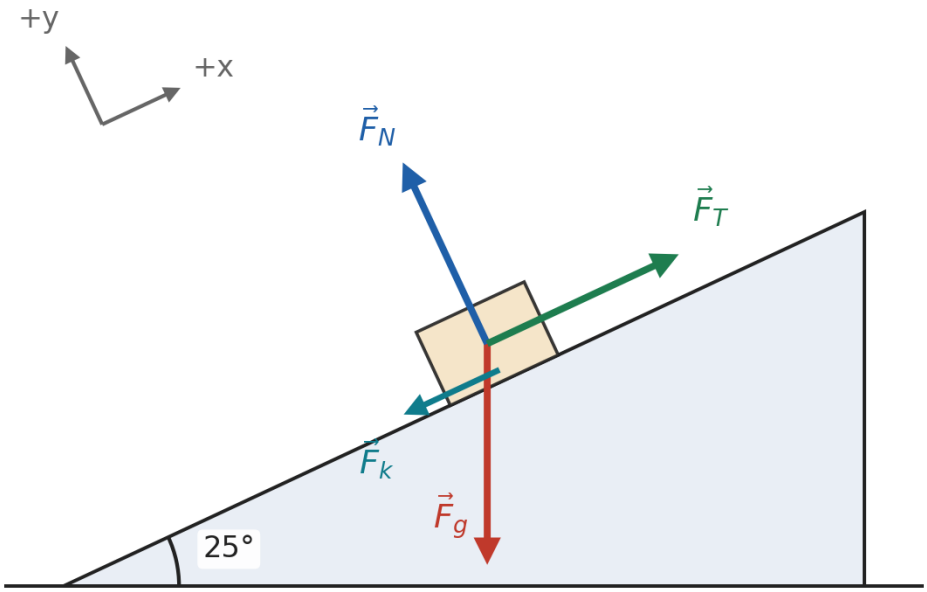
FBD: toboggan



(c) A block getting pulled up a ramp by a string.

Four forces: tension along the string (up the ramp), gravity straight down, the normal force perpendicular to the ramp, and friction down the ramp (opposite the motion). +x is up the ramp.

FBD: block on a ramp



FINAL ANSWER

(a) F_{air} up and F_g down, equal lengths

(b) F_g down, F_N perpendicular to the hill, F_k up the hill

(c) F_T up the ramp, F_g down, F_N perpendicular to the ramp, F_k down the ramp

QUESTION

Static and dynamic equilibrium

Question: Why would both situations be considered “equilibrium”?

Static equilibrium (at rest) and dynamic equilibrium (constant velocity) both have a velocity that is not changing, so a = 0. Newton’s second law then gives a net force of zero:

Formula: $\Sigma\vec{F} = m\vec{a}$

Put in $\vec{a} = 0$: $\Sigma\vec{F} = m(0) = 0$ N

FINAL ANSWER

In both cases a = 0, so the net force is zero. That is what equilibrium means.

EXAMPLE 1

Sled held on an icy hill

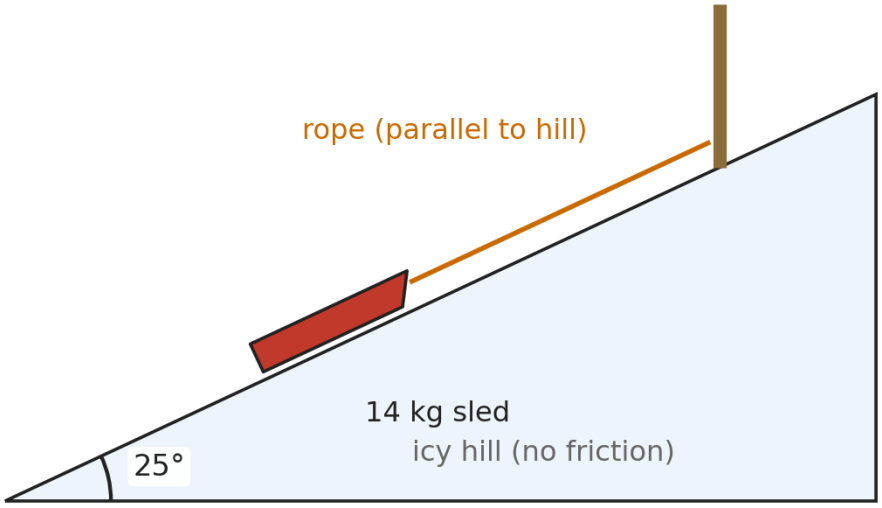
Example 1: A sled has a mass of 14 kg and is on a hill that is inclined 25° to the horizontal, as shown in the figure below. The hill is very icy (friction is negligible), and the sled is held at rest by a rope attached to the post. The rope is parallel to the hill.

(a) Draw the FBD diagram of the scenario.

(b) Calculate the magnitude of the tension in the rope.

(c) Calculate the magnitude of the normal force acting on the sled.

Picture the situation

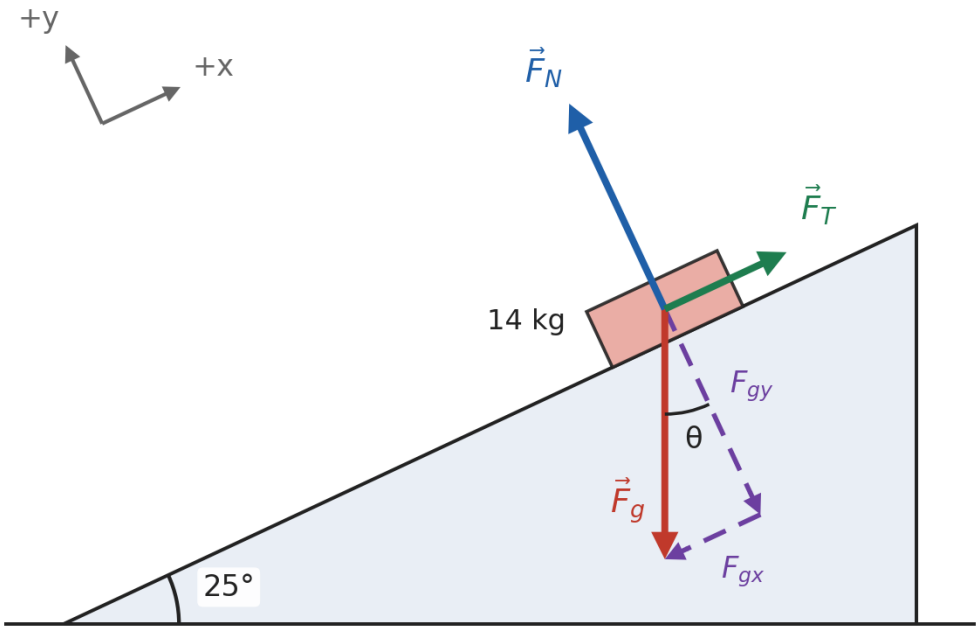


The rope holds the sled still on a frictionless 25° hill.

(a) Draw the FBD diagram of the scenario.

Three forces act on the sled: tension (up the hill, along the rope), the normal force (perpendicular to the hill) and gravity (straight down). Take +x up the hill and +y perpendicular to the hill, away from it. Gravity is the only force not along an axis, so split it into F_{gx} and F_{gy} (dashed).

FBD of the sled



GIVEN

$m = 14$ kg	$\theta = 25°$	$a_x = a_y = 0$ (at rest)
$F_T = ?$	$F_N = ?$	$g = 9.81$ m/s^2

(b) Calculate the magnitude of the tension in the rope.

Step 1 – Find the x component of gravity

F_{gx} is opposite the angle θ in the gravity triangle, so it uses sine. It points down the hill, which is −x:

Formula: $F_{gx} = -mg\sin\theta$

$F_{gx} = -(14)(9.81)\sin 25° = -58.04$ N

Step 2 – Balance the x direction

The sled is at rest, so the x forces add to zero:

Formula: $\Sigma F_x = ma_x$

Put in the forces and $a_x = 0$: $F_T + F_{gx} = 0$

Put in $F_{gx} = -58.04$: $F_T - 58.04 = 0$

Add 58.04 to both sides: $F_T = 58.04$ N

(c) Calculate the magnitude of the normal force acting on the sled.

Step 1 – Find the y component of gravity

F_{gy} is next to the angle θ, so it uses cosine. It points into the hill, which is −y:

Formula: $F_{gy} = -mg\cos\theta$

$F_{gy} = -(14)(9.81)\cos 25° = -124.47$ N

Step 2 – Balance the y direction

Formula: $\Sigma F_y = ma_y$

Put in the forces and $a_y = 0$: $F_N + F_{gy} = 0$

Put in $F_{gy} = -124.47$: $F_N - 124.47 = 0$

Add 124.47 to both sides: $F_N = 124.47$ N

FINAL ANSWER

(a) FBD above: F_T up the hill, F_N perpendicular to the hill, F_g down

(b) $F_T = 58$ N

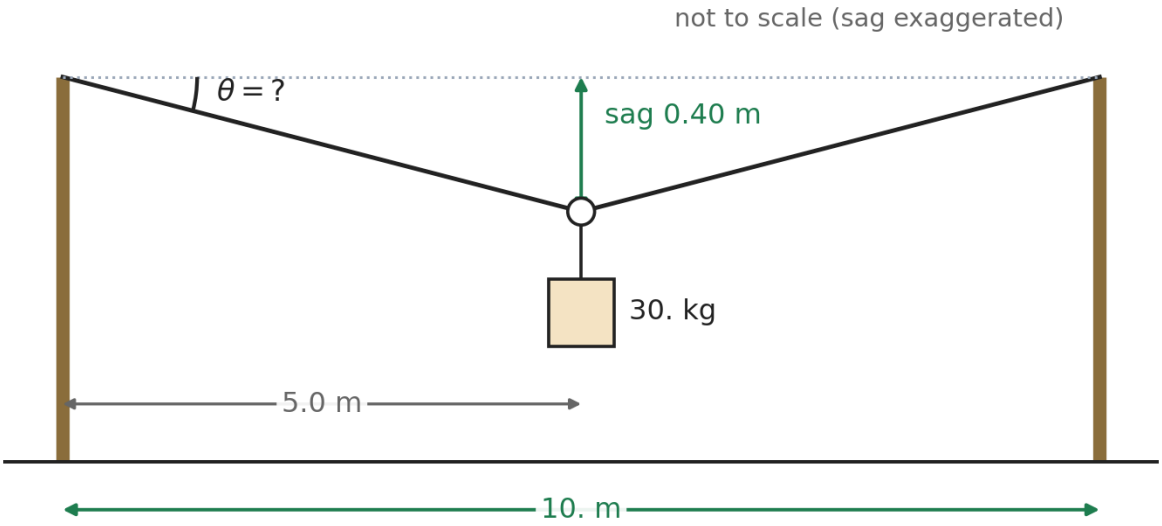
(c) $F_N = 1.2 \times 10^2$ N

EXAMPLE 2

Sagging clothesline

Example 2) A clothesline is attached to high poles 10. m apart. A pulley, allowed to roll freely on the line has a 30. kg mass hanging from it. Find the magnitude of the tension in each half of the clothes-line if the sag at the centre is 0.40 m. (Hint: Draw an FBD of the mass and resolve angled forces into components.) [1.8×10^3 N]

Picture the situation



The pulley sits at the middle, so each half of the line spans 10. m ÷ 2 = 5.0 m sideways and drops 0.40 m.

SOLVING FOR

The tension F_T in each half of the line.

Step 1 – Find the angle of each half

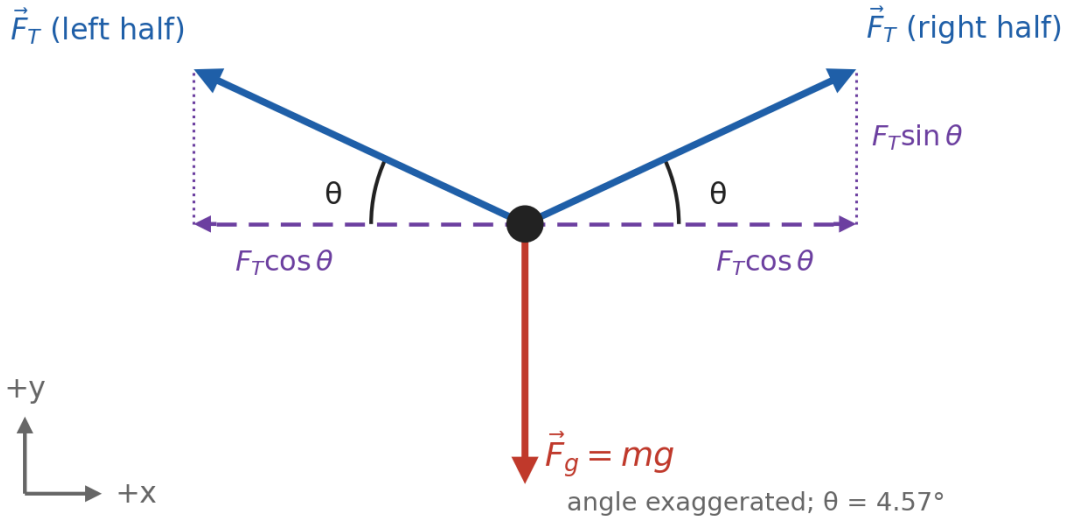
Each half of the line is the long side of a right triangle: 5.0 m across and 0.40 m down. The sag is opposite θ and the 5.0 m is next to it, so use tangent:

$$\text{Formula: } \tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\tan\theta = \frac{0.40\ \text{m}}{5.0\ \text{m}} = 0.080$$

$$\theta = \tan^{-1}(0.080) = 4.57°$$

FBD of the pulley and mass (+x right, +y up)



Step 2 – Look at the x direction

The two halves pull left and right at the same angle. Their x components ($F_T \cos\theta$) cancel only if both halves have the same tension, so one value F_T works for both halves.

Step 3 – Balance the y direction

Each half pulls up with $F_T \sin\theta$. Together they hold up the weight:

Formula: $\Sigma F_y = ma_y$

Put in the forces and $a_y = 0$: $2F_T \sin\theta - mg = 0$

Add mg to both sides: $2F_T \sin\theta = mg$

Divide both sides by $2\sin\theta$: $F_T = \dfrac{mg}{2\sin\theta}$

$$F_T = \frac{(30.)(9.81)}{2\sin 4.57°} = \frac{294.3}{0.1595} = 1845 \text{ N}$$

The tension (1845 N) is about six times the weight (294 N), because only a small part of each pull points up.

FINAL ANSWER

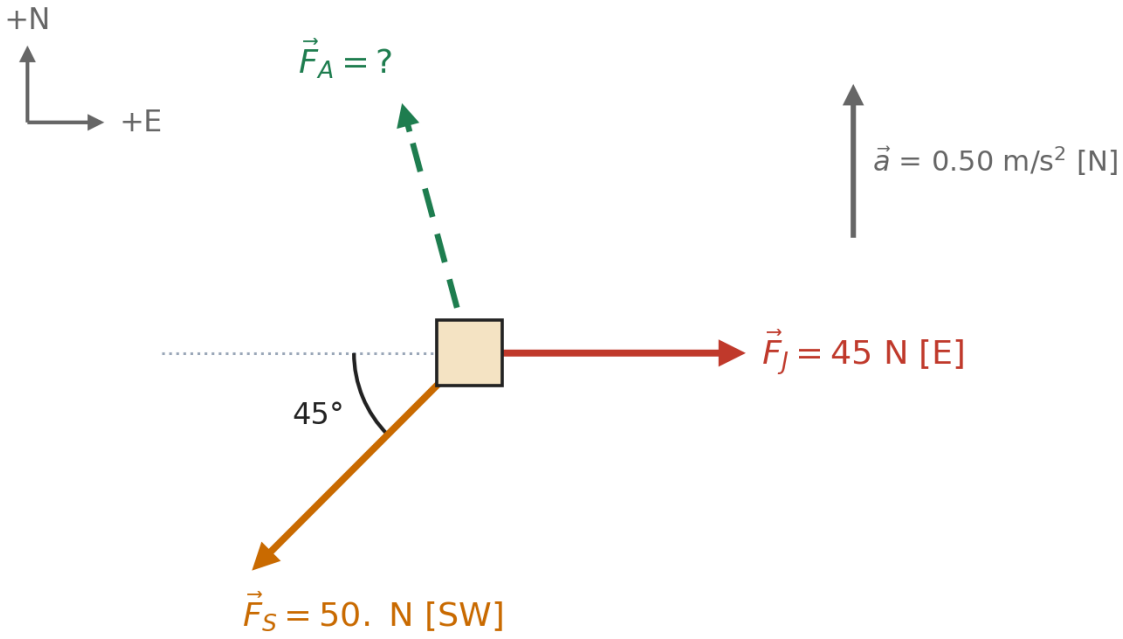
$F_T = 1.8 \times 10^3$ N in each half

EXAMPLE 3

Three students pulling a chair

Example 3) Three students pull horizontally on ropes tied to a 4.0 kg chair, causing it to accelerate at 0.50 m/s^2 [north]. If John pulls with 45 N of force [E] and Saba pulls with 50. N of force [SW], what force is Aleksey exerting? Include an FBD with clearly labelled components.

FBD of the chair, top view (+x east, +y north)



top view (4.0 kg chair)

Gravity and the normal force act up and down, so they don't appear in this top view. They cancel, because the chair doesn't move up or down.

SOLVING FOR

Aleksey's force, F_A (size and direction).

Step 1 – Split Saba's force into components

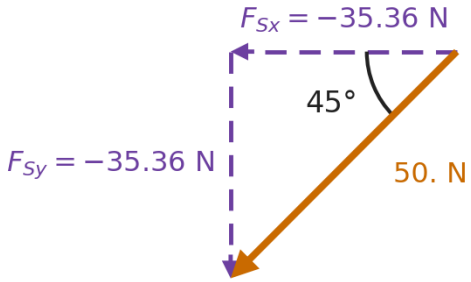
[SW] is exactly 45° between west and south, so both components are the same size. Both point in negative directions (west and south):

Formula: $F_{Sx} = -F_S \cos 45°$ $\quad$ $F_{Sy} = -F_S \sin 45°$

$F_{Sx} = -(50.)\cos 45° = -35.36$ N

$F_{Sy} = -(50.)\sin 45° = -35.36$ N

Diagram: Saba's force and its components



GIVEN

$F_{Jx} = 45$ N	$F_{Jy} = 0$	$m = 4.0$ kg
$F_{Sx} = -35.36$ N	$F_{Sy} = -35.36$ N	$a_x = 0$
$F_{Ax} = ?$	$F_{Ay} = ?$	$a_y = 0.50$ m/s^2

Step 2 – x direction (no acceleration east or west)

Formula: $\Sigma F_x = ma_x$

Put in the forces and $a_x = 0$: $F_{Jx} + F_{Sx} + F_{Ax} = 0$

Put in the numbers: $45 + (-35.36) + F_{Ax} = 0$

Add $45 + (-35.36)$: $9.64 + F_{Ax} = 0$

Subtract 9.64 from both sides: $F_{Ax} = -9.64$ N

Step 3 – y direction (accelerating north)

Formula: $\Sigma F_y = ma_y$

Put in the forces and ma_y: $F_{Jy} + F_{Sy} + F_{Ay} = (4.0)(0.50)$

Put in the numbers: $0 + (-35.36) + F_{Ay} = 2.0$

Add 35.36 to both sides: $F_{Ay} = 37.36$ N

Step 4 – Combine the components

F_{Ax} and F_{Ay} are at right angles, so use the Pythagorean theorem for the size. For the angle, measure from north toward west: the west part (9.64) is opposite the angle and the north part (37.36) is next to it.

Formula: $F_A = \sqrt{F_{Ax}^2 + F_{Ay}^2}$

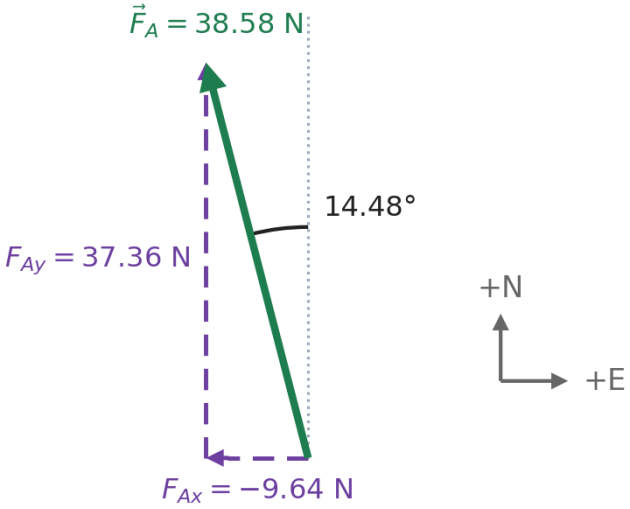
$$F_A = \sqrt{(-9.64)^2 + (37.36)^2} = 38.58 \text{ N}$$

Formula: $\tan\theta = \dfrac{\text{opposite}}{\text{adjacent}} = \dfrac{|F_{Ax}|}{F_{Ay}}$

$$\theta = \tan^{-1}\left(\frac{9.64}{37.36}\right) = 14.48°$$

F_{Ax} is negative (west) and F_{Ay} is positive (north), so the direction is [N 14.48° W]. Rounded to two significant figures, 14.48° becomes 14°.

Diagram: Aleksey's force from its components



FINAL ANSWER

$\vec{F}_A$ = 39 N [N 14° W]